1.

```
sqlite> create table users(id INTEGER primary key autoincrement,name varchar(45), email varchar(45) unique);
sqlite> create table rooms(id INTEGER primary key autoincrement,name varchar(45), capacity int);
sqlite> create table reservations(id integer primary key autoincrement, user_id int, room_id int, date date, foreign key(user_id) references users(id), foreign key(room_id) references rooms(id), UNIQUE(ro
om_id, date ) ON CONFLICT REPLACE);
sqlite> .tables
reservations  rooms         users
```

2.

```
sqlite> insert into users (name, email) values("Raymond", "raymond@ntu.edu.tw");
sqlite> select * from users where name = "Raymond";
1|Raymond|raymond@ntu.edu.tw
```

3.

```
sqlite> insert into users (name, email) values("Mitchell", "mitchell@ntu.edu.tw");
sqlite> insert into users (name, email) values("Annie", "annie@ntu.edu.tw");
sqlite> select * from users where name = "Mitchell" or name = "Annie";
2|Mitchell|mitchell@ntu.edu.tw
3|Annie|annie@ntu.edu.tw
```

4.

```
sqlite> insert into rooms (name, capacity) values("101", 4);
sqlite> insert into rooms (name, capacity) values("102", 6);
sqlite> insert into rooms (name, capacity) values("201", 6);
sqlite> insert into rooms (name, capacity) values("202", 6);
sqlite> insert into rooms (name, capacity) values("301", 12);
```

```
sqlite> select * from rooms;
1|101|4
2|102|6
3|201|6
4|202|6
5|301|12
```

5.

```
sqlite> insert into reservations (user_id, room_id, date) values(1, 5, '2021-12-12');
sqlite> select * from reservations where id = 1;
1|1|5|2021-12-12
```

6.

```
sqlite> insert into reservations (user_id, room_id, date) values(2, 2, '2021-12-12');
sqlite> select * from reservations where id = 2;
2|2|2|2021-12-12
```

7.

```
sqlite> insert into reservations (user_id, room_id, date) values(1, 1, '2021-12-25');
```

```
sqlite> select * from reservations where id = 3;
3|1|1|2021-12-25
```

8.

```
sqlite> insert into reservations (user_id, room_id, date) values(3, 4, '2021-12-25');
sqlite> select * from reservations where id = 4;
4|3|4|2021-12-25
```

9.

```
sqlite> select * from reservations where date = "2021-12-12";
1|1|5|2021-12-12
2|2|2|2021-12-12
```

10.

```
sqlite> select * from reservations where user_id = 1;
1|1|5|2021-12-12
3|1|1|2021-12-25
```

11.

```
sqlite> update reservations set room_id = 3 where id = 3;
sqlite> select * from reservations where id = 3;
3|1|3|2021-12-25
```

12.

```
sqlite> update rooms set capacity = 8 where id = 1;
sqlite> update rooms set capacity = 12 where id = 2;
sqlite> select * from rooms;
1|101|8
2|102|12
```

13.

```
sqlite> select * from reservations where user_id = 1;
1|1|5|2021-12-12
3|1|3|2021-12-25
sqlite> delete from reservations where id = 3;
sqlite> select * from reservations where user_id = 1;
1|1|5|2021-12-12
```